Adarsh Dwivedi

Deep Learning Engineer

+91-9305597756 | 23ucs509@lnmiit.ac.in | linkedin.com/in/adarshdwivedi30 | github.com/adarshcod30 | huggingface.co/adarshcod30

Final-year CS engineer training vision models that state their confidence and abstain when they should. Built a three-backbone deepfake ensemble with calibrated confidence and per-backbone Grad-CAM, and a retinopathy grader using rank-consistent ordinal heads on patient-level splits, with localisation measured against annotated lesions rather than asserted. Models are deployed to Hugging Face and Streamlit, with scope limits published beside the results.

EXPERIENCE

Deloitte India, Digital Excellence Centre **Jun 2026 - Aug 2026**

AI Engineer (Agentic AI) Intern Gurugram, India

- Migrated SAP Agentic, Deloitte's system that automates SAP configuration, from LangGraph running inside a FastAPI request handler onto a self-hosted Aegra agent server, with threads, runs and checkpoints in PostgreSQL inside the firewall at no licence cost.
- Proved the new path on the live SAP tenant: a Bank Accounting goal was planned into 8 configuration tasks, executed in a real browser through Playwright MCP, recovered from a duplicate-entry conflict and a mis-filled field, and finished 8 of 8 in about 15 minutes.
- Built capability in five layers on a multi-agent travel planner, from a single agent to a supervisor with three specialists, a human approval gate, real browser tools over MCP and serving through Aegra, proving each layer before starting the next.

PROJECTS

OpenForensics, Deepfake Face Detection Ensemble | Live | GitHub PyTorch, ResNet50, VGG16, EfficientNetV2, Grad-CAM

- A detector that outputs only a label is unusable as evidence, so built a three-backbone ensemble of ResNet50, VGG16 and EfficientNetV2-B0 with calibrated confidence and an explicit borderline warning band.
- Trained on a two-stage schedule and served per-backbone Grad-CAM, so a reviewer sees which region drove each model's vote instead of trusting the aggregate score.
- Published the training, evaluation and robustness evidence behind every number in an eight-page dashboard, and stated the scope limits plainly: face crops only, swaps rather than generated images, and a single dataset.

Diabetic Retinopathy Screening, Explainable Grading for Rural Screening Programmes | Live | GitHub PyTorch, Ordinal Heads, Grad-CAM, CLAHE

- Grading a 0 as a 4 is not the same mistake as grading a 0 as a 1, so trained ICDR grades 0 to 4 with rank-consistent ordinal heads instead of plain cross-entropy, on patient-level splits where both eyes never straddle train and test.
- Gated every image through a 3-class gradability model fused with focus, illumination and field-of-view features, returning why an image was rejected so the operator can recapture rather than guess.
- Detected sub-pixel microaneurysms with morphological top-hat and matched filtering before CNN classification, and measured Grad-CAM localisation against annotated lesions rather than asserting the heatmaps meant anything.

Vayu, National Air-Quality Intelligence and Enforcement Platform *(Deployed on Google Cloud Run)* **| Live | GitHub** PyTorch, CNN-LSTM, Satellite Imagery

- Trained a CNN-LSTM over a 15,360-cell satellite grid for spatio-temporal air-quality forecasting, fused with LightGBM on ground-station readings so neither source has to be trusted alone.
- Kept the trained network out of the request path, training offline and serving precomputed scores from an embedded DuckDB file, removing model-server latency from every user request.

TECHNICAL SKILLS

Deep Learning: PyTorch, TensorFlow/Keras, CNNs, ResNet50, VGG16, EfficientNetV2, CNN-LSTM, Transformers, Transfer Learning, Ordinal Regression, Grad-CAM, OpenCV

Machine Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, Pandas, NumPy, Feature Engineering, Cross-Validation, Calibration, SHAP, A/B Testing, Statistics

Generative AI: LangGraph, LangChain, RAG, Prompt Engineering, Vector Search, Embeddings, Evaluation and Calibration, Amazon Bedrock, Gemini, OpenAI, Groq, Ollama

Backend and Deployment: FastAPI, Pydantic, Node.js, Express, REST APIs, PostgreSQL, MongoDB, DuckDB, Next.js, React, Streamlit

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL

ACHIEVEMENTS

- SWITCH Energy-X (India): 11th of 93 on the private leaderboard, recovering a hidden physical law from 500,000 unlabeled, 26%-missing sensor rows.
- KSP Datathon 2026 finalist: reached Level 2 with Kadi, a crime analytics platform built end to end over 50 days and deployed on Zoho Catalyst for the Karnataka State Police.

EDUCATION

The LNM Institute of Information Technology, Jaipur **2023 - 2027**

B.Tech, Computer Science and Engineering CGPA: 7.11 / 10

LEADERSHIP

- Coordinator, Debate Society, LNMIIT Jaipur (May 2025 - Apr 2026): led a 150+ member society across debates, discussions and inter-college competitions.
- Secretariat and Organizer, LNMIIT Model United Nations (Aug 2025 - Jan 2026): organised a large-scale MUN conference, coordinating delegates, committees, schedules and logistics.